

# STOCHASTIC PROCESSES & BLACK SCHOLES EQUATION

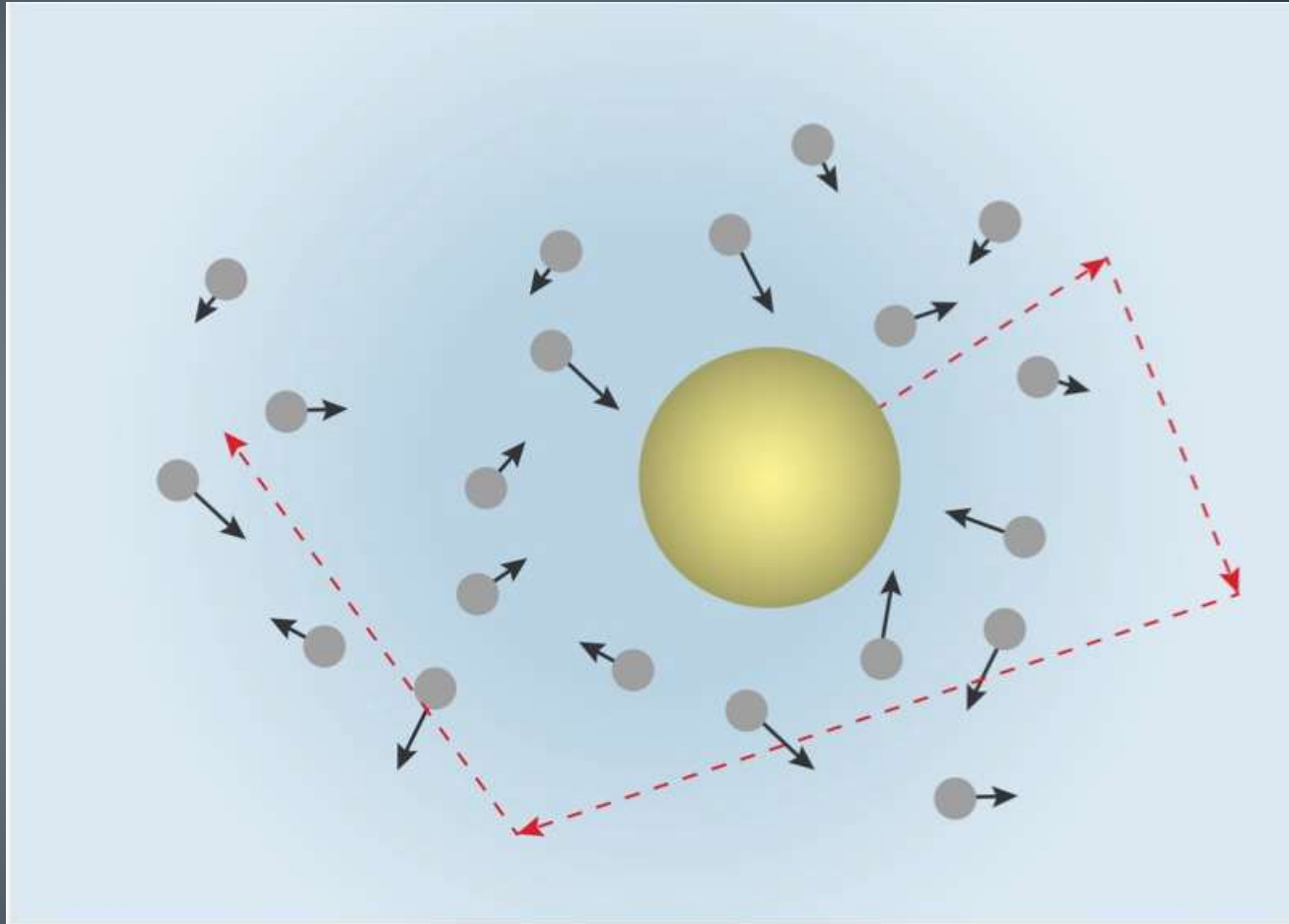
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# Standard Brownian Motion (SBM)

- $W(t, \omega)$  or simply  $W(t)$  is a standard Brownian motion (SBM) if:
  - The mapping  $t \rightarrow W(t)$  is continuous
  - $W(0) = 0$
  - For any  $0 \leq s < t$  the increment  $W(t) - W(s)$  is independent of  $W(\tau)$  for all  $0 \leq \tau \leq s$
  - $W(t) - W(s)$  is a Normal (Gaussian) random variable (r.v.) of mean zero and variance  $t - s$
- Properties of the SBM. If  $W(t)$  is an SBM then:
  - $-W(t)$  is an SBM
  - For any  $c > 0$   $(1/c)W(c^2t)$  is an SBM
  - For  $s \geq 0$  fixed  $W(t + s) - W(s)$  is an SBM

# SBM in more dimensions



<https://scottbembene.com/>

# Itô diffusion

- An Itô diffusion or Itô process  $X(t)$  satisfies the Stochastic Differential Equation (SDE)

$$dX(t) = a(t, X(t))dt + b(t, X(t))dW(t)$$

where

- $a(t, X)$  and  $b(t, X)$  are square integrable (at least locally)
- $W(t)$  is an SBM
- The discrete version of the Itô process (a.k.a. Euler scheme) is

$$X(t + \Delta t) - X(t) = a(t, X(t))\Delta t + b(t, X(t))(W(t + \Delta t) - W(t))$$

- The integral version of the Itô process is

$$X(t) = X(0) + \int_0^t a(s, X(s))ds + \int_0^t b(s, X(s))dW(s)$$

To define the second integral we approximate via step functions and take limits using Lebesgue's Dominated Convergence in  $L^2$

# Intuitions and approximations for $dW$

If  $N$  is a standard Normal r.v. then  $Z(t, \omega) := \sqrt{t}N(\omega)$

- is NOT an SBM
  - No independence of increments
  - Path by path they are not the same a.s.
- But it has the same distribution law i.e.  $\mathcal{N}(0, t)$

Based on the previous approximation, we will use the following intuitions:

- We can assume  $dW \approx \sqrt{dt}$ 
  - This is true in terms of standard deviation because  $E[dW^2] = dt$
- We can simulate Brownian increments as  $W(t + dt) - W(t) \approx \sqrt{dt}N$ 
  - This is the heart of the Euler scheme to simulate SDEs
- In the SDE  $dX = adt + bdW$ 
  - For short times  $0 < t < 1$  the “noise” part  $bdW$  wins
  - For long times  $t > 1$  the “deterministic” part  $adt$  wins

# Itô's lemma 1D

- Let  $X(t)$  be an Itô diffusion i.e.

$$dX = a(t, X)dt + b(t, X)dW$$

and let  $f(t, x)$  be a  $C^2$  function. Then

$$Y(t) := f(t, X(t))$$

is again an Itô process and

$$dY = \frac{\partial f}{\partial t}(t, X)dt + \frac{\partial f}{\partial x}(t, X)dX + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X) \cdot (dX)^2$$

where  $(dX)^2 = dX \cdot dX$  is computed according to the rules

- $dW \cdot dW = dt$
  - $dt \cdot dt = dt \cdot dW = dW \cdot dt = 0$
- If we develop all terms in  $dX$  then (omitting arguments for clarity):

$$dY = \left[ \frac{\partial f}{\partial t} + a \frac{\partial f}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2 f}{\partial x^2} \right] dt + b \frac{\partial f}{\partial x} dW$$

# “Proof” of Itô's lemma 1D

- If  $f(t, x)$  is a  $C^2$  function then its Taylor expansion to second order is

$$\begin{aligned} f(t + dt, x + dX) \approx & f(t, x) + \frac{\partial f}{\partial t}(t, x)dt + \frac{\partial f}{\partial x}(t, x)dX + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, x) \cdot (dX)^2 \\ & + \frac{1}{2} \frac{\partial^2 f}{\partial t^2}(t, x) \cdot (dt)^2 + \frac{\partial^2 f}{\partial t \partial x}(t, x) \cdot dt \cdot dX \end{aligned}$$

- Remember that we use the intuition  $dW \approx \sqrt{dt}$
- In a differential we only keep terms of order  $\leq 1$  in  $dt$ . Therefore:
  - $(dt)^2 = 0$  because it is of order 2
  - $dt \cdot dX = 0$  because it is of order  $\geq 3/2$
  - $(dX)^2$  has only one term of order  $\leq 1$  in  $dt$  namely  $dW \cdot dW = dt$

- Therefore, in differential form the only terms that survive are

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (dX)^2$$

# Geometric Brownian Motion (GBM)

- The Geometric Brownian Motion (GBM) is defined as

$$dX = rXdt + \sigma XdW$$

where  $r$  and  $\sigma$  are constants.

- Observe that assuming GBM prices is equivalent of assuming that the returns follow a Brownian motion with drift:

$$\frac{dX}{X} = rdt + \sigma dW$$

- We have seen that this assumption is not true in reality, but it is widely used because of the nice properties of the Brownian motion
- In order to have more realistic models for prices, we can add more “flavour”
  - Local rate and volatility i.e.  $r = r(t, X)$  and  $\sigma = \sigma(t, X)$
  - Stochastic rate and volatility
    - Ornstein-Uhlenbeck:  $dr = \theta(\mu - r)dt + \gamma dW$
    - Cox-Ingersoll-Ross:  $d\sigma = \theta(\mu - \sigma)dt + \gamma\sqrt{\sigma}dW$



# Explicit solution for the GBM

- Assume a GBM for prices:

$$dX = rXdt + \sigma XdW$$

- Apply Itô's lemma with  $f(t, X) = \log(X)$

$$\begin{aligned} df &= \left[ \frac{\partial f}{\partial t} + a \frac{\partial f}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2 f}{\partial x^2} \right] dt + b \frac{\partial f}{\partial x} dW \\ &= \left[ 0 + rX \frac{1}{X} + \frac{1}{2} \sigma^2 X^2 \left( -\frac{1}{X^2} \right) \right] dt + \sigma X \frac{1}{X} dW \\ &= \left[ r - \frac{1}{2} \sigma^2 \right] dt + \sigma dW \end{aligned}$$

- In integral form we have

$$\log(X(t)) = \log(X(0)) + \left[ r - \frac{1}{2} \sigma^2 \right] t + \sigma W(t)$$

- Taking exponential we finally obtain

$$X(t) = X(0) \exp \left( \left[ r - \frac{1}{2} \sigma^2 \right] t + \sigma W(t) \right)$$

# Black-Scholes equation 1

- Assume the price  $S(t)$  follows a GBM:

$$dS = rSdt + \sigma SdW$$

where  $r$  is the risk-free interest rate, that we assume constant.

- Suppose we have a portfolio consisting on one call option and  $\Delta$  shares on the underlying (a.k.a. the hedge):

$$P(t, S) = C(t, S) - \Delta S$$

- Using Itô's lemma and assuming  $\Delta$  constant we obtain

$$dP = dC - \Delta dS = \frac{\partial C}{\partial t} dt + \left[ \frac{\partial C}{\partial S} - \Delta \right] dS + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} dt$$

- If we choose  $\Delta = \frac{\partial C}{\partial S}$  (a.k.a. delta hedging) then the portfolio is completely deterministic:

$$dP = \left[ \frac{\partial C}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} \right] dt$$

# Black-Scholes equation 2

- On the other hand, the cash value of the deterministic portfolio grows at the risk-free rate:

$$dP = rPdt = r[C - \Delta S]dt$$

- If we assume that there is no arbitrage then both portfolio expressions are the same.
- In consequence, after recalling that  $\Delta = \frac{\partial C}{\partial S}$  it follows that:

$$\left[ \frac{\partial C}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} \right] dt = r \left[ C - S \frac{\partial C}{\partial S} \right] dt$$

- Eliminating the  $dt$  and regrouping we obtain the Black-Scholes equation:

$$\frac{\partial C}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS \frac{\partial C}{\partial S} - rC = 0$$

- This equation has a terminal condition at maturity time  $T > 0$  i.e. the payoff of a call option of strike  $K$ :

$$C(T, S) = \max(S - K, 0)$$

# Black-Scholes and the heat equation

- After performing some hardcore changes of variables, the Black-Scholes equation can be transformed into the heat equation

$$\frac{\partial u}{\partial \tau} - \frac{\partial^2 u}{\partial z^2} = 0$$

with a certain initial condition  $u(0, z) = u_0(z)$

- If the initial condition  $u_0(z)$  is nice enough then there is a unique solution:

$$u(\tau, z) = \int_{-\infty}^{+\infty} u_0(y) K(\tau, z, y) dy$$

where  $K(\tau, z, y)$  is the Heat Kernel defined as

$$K(\tau, z, y) := \frac{1}{2\sqrt{\pi\tau}} \exp\left(-\frac{(z-y)^2}{4\tau}\right)$$

- If you are courageous enough, you could prove by brute force that the heat kernel solves the heat equation

# Solution to the Black-Scholes equation

- From the shape of the Heat Kernel  $K(\tau, z, y)$  we can see that it is related to a Gaussian or Normal density function of mean  $z$  and variance  $t$
- Therefore, it should not be a surprise that the exact solution to the Black-Scholes equation is related to the standard normal distribution
- The value of a call option in terms of its parameters is:

$$C(t, S) = SN(d_1) - Ke^{-r(T-t)}N(d_2)$$

where

$$d_1 = \frac{1}{\sigma\sqrt{T-t}} \left[ \log\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t) \right]$$
$$d_2 = d_1 - \sigma\sqrt{T-t}$$

and  $N(\cdot)$  is the cumulative distribution function of the standard Normal r.v.

Thank you for your  
attention

